

KELELA SOLUTIONS

Cybersecurity Compliance Platform

Continuous compliance for INSA, NBE, and international security frameworks — built for Ethiopian banks, telecoms, and critical infrastructure

PLATFORM OVERVIEW · ARCHITECTURE · SECURITY · ROADMAP

THE PROBLEM

Compliance is mandatory. Proving it is where organizations struggle.

Fragmented obligations

Ethiopian banks and telecoms must satisfy INSA and NBE requirements in parallel, often proving the same underlying control twice in different formats.

Spreadsheet-based tracking

Without a system of record, ownership, evidence, and deadlines live in disconnected spreadsheets that nobody fully trusts.

Time-critical obligations missed

INSA requires incident disclosure within 48 hours. Without an active countdown, teams discover a missed deadline only after it matters.

No audit-ready evidence trail

When an auditor asks 'prove it,' evidence is scattered, undated, and impossible to verify as untampered.

THE OPPORTUNITY

There is no dedicated compliance platform serving this market today

INSA builds security tooling. General software firms build custom one-off systems. Global GRC platforms (Drata, Sprinto, OneTrust) don't understand INSA or NBE directives at all.

12

Sectors now under INSA audit power, per the current proclamation

18

Distinct compliance obligations enterprises must satisfy

48h

Mandatory INSA incident disclosure window



Regulators are actively converging

NBE and INSA are working to harmonize their directives into a unified, risk-based national framework. Whoever builds the system of record for that transition earns the default vendor position with every regulated enterprise in the country.

PURPOSE

What Kelela Solutions is — and isn't



Software vendor

We build and operate the platform: the control catalog, the evidence vault, the incident tracker — maintained centrally, shipped to every client at once.



Compliance guidance provider

Raw regulatory text becomes a trackable, owned, evidence-backed control. That translation work is expertise, not just software.



Platform operator, not the compliant party

The client remains legally responsible for their own compliance. We provide the system of record — we don't file on their behalf or assume their liability.

One line: Kelela Solutions doesn't make companies compliant — it makes compliance something they can actually manage and prove.

HOW IT'S BUILT

A two-layer, multi-tenant architecture

Platform Console · operated by Kelela Solutions staff

Super Admin (1) and Platform Admins (several) onboard tenants, oversee aggregate health across every organization, and never casually browse one tenant's raw compliance data.



Kelela Solutions (internal)

Uses its own product

Abyssinia Digital Bank S.C.

Financial · Trial

Next client organization…

Onboarded in minutes

Every tenant runs identical code — same UI, same features. What differs is the data, walled off by organization ID on every query.

WHAT IT DOES

Core platform features

Compliance Control Matrix

Every requirement tracked with owner, status, risk level, and due date across 6 frameworks.

Live Incident SLA Tracking

48-hour INSA disclosure countdown, with auto-flagging when a linked control breaches.

Tamper-Evident Evidence Vault

Every uploaded file is SHA-256 hashed on receipt and re-verified on every download.

One-Click Audit Reports

Executive and full auditor-ledger PDF exports, generated server-side on demand.

Readiness Trends & Charts

30-day readiness history, framework status breakdowns, and risk distribution at a glance.

Team & Tenant Management

Org admins invite their own team; Kelela staff onboard and oversee tenants centrally.

COVERAGE

Six frameworks, independently tracked

INSA

Critical Infrastructure Proclamation

Mandatory · Ethiopia

NBE

Cybersecurity Directive

Mandatory · Financial sector

ISO 27001

Information Security Management

International baseline

NIST CSF 2.0

Govern · Identify · Protect · Detect · Respond · Recover

International

PCI DSS v4.0

Cardholder Data Security

Conditional · Payment data

SOC 2

Trust Services Criteria

Enterprise due diligence

Controls that satisfy the same underlying requirement across frameworks are cross-referenced — proving one satisfies several.

ACCESS MODEL

Three tiers, two scopes, zero ambiguity



Super Admin

Platform-wide · 1

Full control: onboard or remove tenants, manage Platform Admins, suspend access.



Platform Admin

Platform-wide · several

Onboard and view tenants day-to-day — cannot suspend a tenant or manage other staff.



Org Admin & Team

One tenant org · many

Full control within their own organization — invite teammates, assign controls, manage incidents.

Enforced with dedicated runtime guards — not just UI conditionals — on every route.

SECURITY

Built security-first, not security-added

Argon2id password hashing

OWASP-recommended memory-hard hashing, not a fast general-purpose hash.

TOTP multi-factor authentication

QR-code enrollment; mandatory platform-wide for Kelela staff, org-toggleable for tenants.

Immutable audit log

Every status change, upload, and export recorded — who, what, and when, append-only.

SHA-256 evidence integrity

Every file hashed on receipt and re-verified on download; tampering is detected, not assumed absent.

Brute-force lockout

Accounts lock after 5 failed attempts; disabled accounts can't self-reactivate via password reset.

Security response headers

HSTS, X-Frame-Options, and a locked-down Permissions-Policy on every response.

HOW IT'S BUILT

Technology stack

Frontend	Next.js 14 (App Router) · TypeScript · Tailwind CSS · Recharts
Backend	Next.js Route Handlers · Zod validation at every boundary
Database	PostgreSQL · Prisma ORM · audit-log tables with no update/delete grant
Auth & Security	JWT in httpOnly cookies · Argon2id · TOTP MFA · RBAC guards
Email & Storage	SMTP with local-dev fallback · evidence storage abstracted for S3 swap-in

Chosen for one reason: every piece is something a small team can operate and reason about end to end — no unnecessary infrastructure.

WHAT'S NEXT

Roadmap

SHIPPED

- Team invites & self-service password reset
- Mandatory MFA policy, per-org and platform-wide
- Soft-delete tenant offboarding with full audit trail

PLANNED

Sector-based auto control provisioning

A telecom shouldn't see PCI DSS controls; auto-populate by sector at onboarding.

Production cloud evidence storage

Swap local-disk storage for S3/GCS — the interface is already written for it.

Automated SLA & deadline alerts

Email/SMS notification the moment an incident clock or control due-date is at risk.

Amharic language support

Compliance officers working day-to-day shouldn't be limited to English.

Mobile companion app

Push alerts for incident SLAs and approvals while away from a desk.

SIEM & ticketing integrations

Pull incidents in from existing security tooling instead of manual re-entry.



Compliance that can be managed — and proven.

Kelela Solutions · Cybersecurity Compliance Platform

Thank you